

E.G.S. PILLAY ENGINEERING COLLEGE

(Autonomous)

NAGAPATTINAM – 611 002.

(Affiliated to Anna University, Chennai | Accredited by NAAC with ‘A++’ Grade

Accredited by NBA | Approved by AICTE, New Delhi)



B.E. BIOMEDICAL ENGINEERING (R–2023)

SECOND YEAR: THIRD SEMESTER

COURSE CODE	COURSE NAME	CATEGORY	L	T	P	C	MAX. MARKS		
							CA	ES	TOTAL
THEORY COURSES									
2301MA302	Probability and Stochastic Processes	BSC	3	1	0	4	40	60	100
2302BM301	Biosciences in Medical Engineering	PCC	3	0	2	4	50	50	100
2302BM302	Bio Process Control Systems	PCC	3	0	0	3	40	60	100
2302BM303	Bio medical Circuits and Networks	PCC	3	0	2	3	40	60	100
2302BM304	Human Anatomy and Physiology	PCC	3	0	2	3	40	60	100
2301GEX07	Environmental Science and Sustainability	BSC	1	0	2	2	40	60	100
LABORATORY COURSES									
2302BM351	Human Anatomy and Physiology Laboratory	PCC	0	0	2	1	60	40	100
2302BM352	Circuits and networks Laboratory	PCC	0	0	2	1	60	40	100
2304GE301	Professional Development Course - 1	EEC	0	0	2	1	100	--	100
TOTAL			16	1	14	22	470	430	900

2301MA302	PROBABILITY THEORY AND STOCHASTIC PROCESSES										L	T	P	C
	(B.E-ECE and BME)										3	1	0	4
PREREQUISITE:Maths- 1,Maths- 2														
	1.Advanced multivariate differential calculus and integral calculus													
	2.Matrices													
COURSE OBJECTIVES:														
1. To analyze the concepts of probability, random variables and distribution functions.														
2. To acquire skill in handling situation with more than one random variable with time function.														
3. To analyze the concepts of Covariance and correlation.														
4. To analyze the concept of Marko v process.														
5. To acquire skills in Linear systems with random inputs.														
COURSE OUTCOMES:														
On the successful completion of the course, students will be able to														
CO1:	Apply probability techniques to analyze the performance of Electronic systems.(K3)													
CO2:	Apply standard distributions in describing real life phenomena.(K3)													
CO3:	Solve problems involving more than one random variable.(K3).													
CO4:	Make use of theorems related to random signals(K3)													
CO5:	Interpret the response of random input to linear time invariant systems.(K3)													
COs Vs POs MAPPING:														
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
	CO1	3	2	1	-	-	-	-	-	-	-	1	1	
	CO2	3	2	1	-	-	-	-	-	-	-	1	1	
	CO3	3	2	1	-	-	-	-	-	-	-	1	1	
	CO4	3	2	1	-	-	-	-	-	-	-	1	1	
	CO5	3	2	1	-	-	-	-	-	-	-	1	1	
COs Vs PSOs MAPPING:														
	COs	PSO1	PSO2	PSO3										
	CO1	1	1	-										
	CO2	1	1	-										
	CO3	1	1	-										
	CO4	1	1	-										
	CO5	1	1	-										
COURSE CONTENTS:														
Module I	PROBABILITY THEORY											9+3 Hours		
Sets and set operations -Probability- Conditional probability and Bayes theorem-Discrete and continuous random variables – Moments – Moment generating functions –Real Time Problems.														

Module II	ONE -DIMENSIOAL RANDOM VARIABLE	9+3 Hours
Discrete Distributions: Binomial, Poisson, Geometric - Continuous Distributions: Uniform, Exponential, Normal distributions- Applications of Distribution in Engineering Problems		
Module III	TWO - DIMENSIONAL RANDOM VARIABLES	9+3 Hours
Joint distributions – Marginal and conditional distributions – Covariance – Correlation and Linear regression Transformation of random variables.		
Module IV	STOCHASTIC PROCESSES	9+3 Hours
Classification -Stationary process – Markov process – transition probabilities – Limiting distributions – Poisson process		
Module V	SPECTRAL DENSITIES AND LINEAR SYSTEMS WITH RANDOM INPUTS	9+3 Hours
Auto correlation function - cross correlation function – Properties - power spectral density-cross spectral density Properties - Linear time invariant system- system transfer function-Linear system with random inputs-Auto correlation and cross correlation functions of input and output.		
TOTAL:45+15=60 HOURS		
REFERENCES:		
H. Stark and J. Woods, Probability and Random Processes with Applications to Signal Processing, Third Edition, Pearson Education		
2. A. Papoulis and S. Unnikrishnan Pillai, Probability, Random Variables and Stochastic Processes, Fourth Edition, McGraw Hill..		
3. K. L. Chung, Introduction to Probability Theory with Stochastic Processes, Springer International.		
4. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability, UBS Publishers		
5. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Stochastic Processes, UBS Publishers.		
6. S. Ross, Introduction to Stochastic Models, Harcourt Asia, Academic Press		

2302BM301	BIOSCIENCES IN MEDICAL ENGINEERING										L	T	P	C
	(B.E- BME)										3	0	2	4
PREREQUISITE :														
	Basic knowledge of chemistry													
COURSE OBJECTIVES:														
1. To study structural and functional properties of carbohydrates, proteins, lipids and nucleic acids														
2. To emphasize structural and functional aspects of living organisms														
3. To discuss the concepts of microbial culture and immunopathology														
COURSE OUTCOMES:														
On the successful completion of the course, students will be able to														
CO1:	Understand the fundamental concepts of biochemistry correlated to biomedical engineering													
CO2:	Outline the concept of cell degeneration, cell repair and Neoplasia.													
CO3:	Analyze the effect of derangement in haematology													
CO4:	Analyze the growth of bacterial cultures													
CO5:	Analyze the pathological conditions related to immunity													
COs Vs POs MAPPING:														
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
	CO1	3	-	-	-	-	-	-	-	-	-	-	2	
	CO2	3	-	-	-	2	-	-	-	-	-	-	2	
	CO3	3	2	-	-	2	-	-	-	-	-	-	2	
	CO4	3	2	-	-	2	-	-	-	-	-	-	2	
	CO5	3	2	-	-	2	-	-	-	-	-	-	2	
COs Vs PSOs MAPPING:														
		COs	PSO1	PSO2	PSO3									
		CO1	1	-	-									
		CO2	1	-	-									
		CO3	1	-	-									
		CO4	1	-	-									
		CO5	1	-	-									
COURSE CONTENTS:														
Module I	INTRODUCTION TO BIOCHEMISTRY												9 Hours	
Introduction to Biochemistry, water as a biological solvent, weak acid and bases, pH, buffers, Handerson-Hassel balch equation, physiological buffers, fitness of the aqueous environment for living organism. Principle of viscosity, surface tension, adsorption, diffusion, osmosis and their applications in biological systems. Introduction to Biomolecules, Biological membrane, Clinical application of Electrolytes and radioisotopes														
Module II	CELL DEGENERATION, REPAIR AND NEOPLASIA												9 Hours	

Cell injury - Reversible cell injury and irreversible cell injury and Necrosis, Apoptosis, Intracellular accumulations, Pathological Calcification-Dystrophic and Metastatic. cellular adaptations of growth and differentiation, Inflammation and Repair including fracture healing, Neoplasia, Classification, Benign and malignant tumors, carcinogenesis, spread of tumours, autopsy and biopsy.		
Module III	FLUID AND HEMODYNAMIC DERANGEMENTS	9 Hours
Edema, Hyperemia/Ischemia, normal haemostasis, thrombosis, disseminated intravascular coagulation, embolism, infarction, shock, chronic venous congestion. Haematological disorders- Bleeding disorders, Leukaemia, Lymphomas Haemorrhage.		
Module IV	MICROBIAL CULTURES	9 Hours
Morphological features and structural organization of bacteria, growth curve, identification of bacteria, culture media and its types, culture techniques and observation of culture.		
Module V	IMMUNOPATHOLOGY	9 Hours
Natural and artificial immunity, types of Hypersensitivity, antibody and cell mediated tissue injury: opsonisation, phagocytosis, inflammation, Secondary immune deficiency including HIV infection. Auto-immune disorders: Basic concepts and classification, SLE. Antibodies and its types, antigen and antibody reactions, immunological techniques: immune diffusion, immune - electrophoresis, RIA and ELISA, monoclonal antibodies.		
		TOTAL: 45 Hours
REFERENCES:		
1.Rafi MD, "Textbook of Biochemistry for Medical Student", Second Edition, University Press, 2014		
2.Donold Voet, Judith, G.Voet and Charlotte W. Pratt, "Principles of Biochemistry", 4th Edition, John Wiley and Sons, New Delhi, 2012.		
3.David.W.Martin, Peter.A.Mayes, Victor. W.Rodwell, Harper's Illustrated Biochemistry, LANGE Medical Publications, 2018		
4.Keith Wilson, John Walker, Practical Biochemistry- Principles & Techniques, Oxford University Press,2009		
5.Harsh Mohan, Text book of Pathology, Jaypee Brothers Medical publishers private Limited, 8th Edition,2019		

2302BM302	BIO PROCESS CONTROL SYSTEMS									L	T	P	C
	(B.E-BME)									3	0	0	3
PREREQUISITE:													
	1. Linear Algebra, Calculus and Partial Differentiation, Calculus, Ordinary Differential Equations And Complex Variables 3.Basic knowledge of biology,												
COURSE OBJECTIVES:													
1. To develop the system modelling and various methods of representation.													
2. To implement the time response analysis of a system													
3. To employ the frequency response analysis of a system													
4. To comprehend the biological control system.													
5. To develop the models based on biological systems													
COURSE OUTCOMES:													
On the successful completion of the course the students will be able to,													
CO1:	Determine the mathematical model of the physical system.												
CO2:	2. Calculate the transient and steady State behaviour of systems using standard test signals.												
CO3:	3. Determine the frequency response of systems using different charts.												
CO4:	4. Describe the biological control systems												
CO5:	5. Express the applications control system in biomedical field.												
COs Vs POs MAPPING:													
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
	CO1	3	2	2	1	1	-	-	-	-	-	1	1
	CO2	3	2	2	1	1	-	-	-	-	-	1	1
	CO3	3	2	2	1	1	-	-	-	-	-	1	1
	CO4	3	2	2	1	1	-	-	-	-	-	1	1
	CO5	3	2	2	1	1	-	-	-	-	-	1	1
COs Vs PSOs MAPPING:													
	COs	PSO1	PSO2	PSO3									
	CO1	1	1	-									
	CO2	1	1	-									
	CO3	1	1	-									
	CO4	1	1	-									
	CO5	1	1	-									
COURSE CONTENTS:													
Module I	INTRODUCTION TO CONTROL SYSTEM												9 Hours
Open and Closed loop Systems, Mathematical Modelling of systems, Block diagram and signal flow graph representation of systems - reduction of block diagram and signal flow graph.													
Module II	TIME RESPONSE ANALYSIS												9 Hours

Time response, Time domain specifications, Types of test input, I and I order system response, Error coefficients, Generalized error series - Steady state error		
Module III	FREQUENCY RESPONSE ANALYSIS	9 Hours
Frequency response, Bode plot, Polar plot, Determination of closed loop response from openloop response, Correlation between frequency domain and time domain specifications.		
Module IV	BIOLOGICAL CONTROL SYSTEM	9 Hours
Introduction to Physiological control systems- Illustration, Linear models of physiological systems, Difference between engineering and physiological control systems. Simple models of muscle stretch reflex action		
Module V	BIOMEDICAL APPLICATIONS	9 Hours
Examples of Biological control Systems: Cardiovascular Control System, Endocrine Control Systems, Pupil Control System, Skeletal Muscle Servomechanism, Oculo - motor system, sugar level Control Mechanism. Temperature control, Blood pressure control.		
TOTAL: 45 Hours		
Further readings:		
Test book:		
1.A. Nagoorkani ,Control Systems ,Linear Control Theory Book Control System Engineering		
2. Michael C. K. Khoo Biomedical Engineering Department University of Southern California ,Physiological Control Systems Analysis, Simulation, And Estimation, IEEE Engineering in Medicine and Biology Society, Sponsor, A Johnwiley & Sons, INC.,Publication		
3.Control Systems Engineering - I. J. Nagrath and M. Gopal, New Age International (P) Limited, Publishers.		
REFERENCES:		
1. Control Systems Theory and Applications - S. K. Bhattacharya, Pearson.		
2. Control Systems Engineering - S. Palani, TMH		
. 3. Control Systems - N. K. Sinha, New Age International (P) Limited Publishers.		
4. Control Systems by S.Hasan Saeed, KATSON BOOKS.		
5. Solutions and Problems of Control Systems by A.K. Jairath, CBS Publishers.		

2302BM303	BIO MEDICAL CIRCUITS AND NETWORKS										L	T	P	C
	(B.E-BME)										3	0	2	3
PREREQUISITE:														
1.Semiconductor physics														
COURSE OBJECTIVES:														
1. To introduce the basic concepts of DC electrical circuits and analysis.														
2. To introduce different methods of circuit analysis using Network theorems.														
3. To understand resonance and coupled circuits.														
4. To study the transient and steady state response of the circuits.														
5. To discuss the electrical network graphs.														
COURSE OUTCOMES:														
On the successful completion of the course students will be able to,														
CO1:	Calculate the circuit parameters using mesh and nodal method													
CO2:	Apply various network theorems for the analysis of electrical circuits.													
CO3:	Examine the concept of resonance and coupled circuits													
CO4:	Determine transient response of DC circuits.													
CO5:	Understand the graphs of networks													
COs Vs POs MAPPING:														
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
	CO1	3	2	2	1	1	-	-	-	1	-	1	1	
	CO2	3	2	2	1	1	-	-	-	1	-	1	1	
	CO3	3	2	2	1	1	-	-	-	1	-	1	1	
	CO4	3	2	2	1	1	-	-	-	1	-	1	1	
	CO5	3	2	2	1	1	-	-	-	1	-	1	1	
COs Vs PSOs MAPPING:														
	COs	PSO1	PSO2	PSO3										
	CO1	1	1	-										
	CO2	1	1	-										
	CO3	1	1	-										
	CO4	1	1	-										
	CO5	1	1	-										
COURSE CONTENTS:														
Module I	BASIC CIRCUIT ANALYSIS												9 Hours	
Resistive elements - Ohm's Law- Resistors in series and parallel circuits-voltage to current and current to voltage source conversion– Kirchhoff's laws – Mesh current and node voltage methods of analysis-Solution of mesh equations by Cramer's rule														
Module II	NETWORK REDUCTION AND THEOREMS FOR DC CIRCUITS												9 Hours	
Network reduction: voltage and current division-star delta conversion-Thevenin's' and Norton Theorems – Superposition Theorem – Maximum power transfer theorem														

Module III	RESONANT CIRCUITS	9 Hours
Series and parallel resonance – their frequency response – Quality factor and Bandwidth – concept of Self and mutual inductance – Coefficient of coupling-Dot rule convention in Coupled circuits		
Module IV	TRANSIENTS FOR DC CIRCUITS	9 Hours
Natural response-Forced response -Transient response of RL, RC and RLC Circuits with dc excitation -Solution of Problems in Circuit Transients		
Module V	ELECTRICAL NETWORKS	9 Hours
Graph of Network: Concept of Tree Branch, Tree link, junctions, Incident matrix, Tie-set matrix, Cutset matrix, determination of loop current and node voltages.		
TOTAL: 45 Hours		
REFERENCES :		
1.William F.Ganong, "Review of Medical Physiology", 26nd Edition, McGraw Hill, New Delhi,2019		
2.Boylestad Robert L., Nashelsky Louis, Electronics Devices & Circuits, Pearson Education.		
3.Circuits and Networks., U.A.Bakshi, A.V.Bakshi,Technical publications,2011.		

2302BM304	HUMAN ANATOMY AND PHYSIOLOGY										L	T	P	C
	(B.E-BME)										3	0	2	3
PREREQUISITE:														
	Basic knowledge of biology													
COURSE OBJECTIVES:														
1. To identify all the organelles of an animal cell and their function														
2. To understand structure and functions of the various types of systems of human body														
3. To demonstrate their knowledge of importance of anatomical features and physiology of human systems														
COURSE OUTCOMES:														
On the successful completion of the course students will be able to,														
CO1:	Understand the structure and function of cells and tissues.													
CO2:	Outline Skeletal, Muscular and Respiratory systems and their importance.													
CO3:	Interpret the working of Cardiovascular and Lymphatic systems.													
CO4:	Analyze the functions of Nervous, Endocrine and Special sensory systems.													
CO5:	Analyze the functions of various components of the digestive system and evaluate the role of the urinary system in maintaining homeostasis.													
COs Vs POs MAPPING:														
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	
	CO1	3	-	-	-	2	-	-	-	-	-	1	1	
	CO2	2	-	-	-	2	-	-	-	-	-	1	1	
	CO3	3	-	-	1	2	-	-	-	-	-	1	1	
	CO4	3	2	-	1	2	-	-	-	-	-	1	1	
	CO5	3	2	-	1	2	-	-	-	-	-	1	1	
COs Vs PSOs MAPPING:														
		COs	PSO1	PSO2	PSO3									
		CO1	1	-	-									
		CO2	1	-	-									
		CO3	1	-	-									
		CO4	1	-	-									
		CO5	1	-	-									
COURSE CONTENTS:														
Module I	BASIC ELEMENTS OF HUMAN BODY												9 Hours	
Cell: Structure and organelles - Functions of each component in the cell. Cell membrane - transport across cell membrane - origin of cell membrane potential - Action potential - Cell to cell signalling- Cell Division. Tissue Types - Specialized tissues - functions.														
Module II	SKELETAL, MUSCULAR AND RESPIRATORY SYSTEMS												9 Hours	

Skeletal: Bone types and functions - Joint-Types of Cartilage and functions. Muscular: Parts of Muscle-Movements. Respiratory: Components of respiratory system-Respiratory Mechanism. Types of respiration - Oxygen and carbon dioxide transport and acid base regulation.		
Module III	CARDIOVASCULAR AND LYMPHATIC SYSTEM	9 Hours
Cardiovascular: Blood composition - functions of blood - functions of RBC. WBC types and their functions. Blood groups - Importance of blood groups - Identification of blood groups. Structure of heart - Conducting system of heart - Properties of Cardiac muscle - Cardiac cycle - Types of Blood Vessel - ECG - Heart sound - Volume and pressure changes and regulation of heart rate - Coronary Circulation. Factors regulating Blood flow. Lymphatic: Parts and Functions of Lymphatic systems - Types of Lymphatic organs and vessels.		
Module IV	NERVOUS, ENDOCRINE AND SPECIAL SENSORY SYSTEMS	9 Hours
Nervous: Structure of Neuron - Types of Neurons. Synapses and types. Condition of action potential in neuron. Brain - Divisions of brain lobes - Cortical localizations and functions - EEG. Spinal cord - Tracts of spinal cord - Reflex mechanism - Types of reflex. Autonomic nervous system and its functions. Endocrine - Pituitary and thyroid gland. Special sensory: Optics of Eye - Retina - Photochemistry of Vision - Neurophysiology of Vision - EOG. Structure and functions of Internal Ear Mechanism of Hearing - Auditory pathway, Hearing Tests.		
Module V	DIGESTIVE AND URINARY SYSTEMS	9 Hours
Digestive: Organs of Digestive system - Digestion and Absorption. Urinary: Structure of Kidney and Nephron. Mechanism of Urine formation and acid base regulation - Urinary reflex - Homoeostasis and blood pressure regulation by urinary system.		
TOTAL: 45 Hours		
REFERENCES :		
1. William F. Ganong, "Review of Medical Physiology", 26 nd Edition, Mc Graw Hill, New Delhi, 2019		
2. Gillian Pocock, Christopher D. Richards, "The Human Body - An Introduction for Biomedical and Health Sciences", Oxford University Press, USA, 2009		
3. Eldra Pearl Solomon, "Introduction to Human Anatomy and Physiology", W.B. Saunders Company, Harcourt Brace Jovanovich, 2003		
4. Frank H. Netter, "Atlas of human anatomy", Netter basic science, 7 th edition 2019.		

2301GEX07	ENVIRONMENTAL SCIENCE AND SUSTAINABILITY								L	T	P	C	
	(Common to all Branches of B.E/ B.Tech)								1	0	2	2	
PREREQUISITE :													
	Basic knowledge of eco system												
COURSE OBJECTIVES:													
1. Realize the interdisciplinary and holistic nature of the environment.													
2. Understand how natural resources and environment affect the quality of life and stimulate the quest for sustainable development.													
COURSE OUTCOMES:													
At the end of the course the student will be able to													
CO1:	Describe the importance of ecosystem.												
CO2:	Describe the various environmental issues and its prevention.												
CO3:	Organize various natural resources and the immediate need to conserve it.												
CO4:	Select the various ways of conservation of biodiversity.												
CO5:	Investigate the different types of pollution and its effects												
COs Vs POs MAPPING:													
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
	CO1	2	1	-	-	-	-	3	-	-	-	-	-
	CO2	2	1	-	-	-	-	3	-	-	-	-	-
	CO3	3	2	1	1	1	1	3	2	2	2	3	-
	CO4	3	2	1	1	1	1	3	2	2	2	3	-
	CO5	3	2	1	1	1	1	3	2	2	2	3	-
COs Vs PSOs MAPPING:													
	COs	PSO1	PSO2	PSO3									
	CO1	-	-	-									
	CO2	-	-	-									
	CO3	-	-	-									
	CO4	-	-	-									
	CO5	-	-	-									
COURSE CONTENTS:													
Module I	ECO SYSTEM										8 Hours		
Concept of an ecosystem – structure and function of an ecosystem – producers, consumers and decomposers. Introduction, types, characteristic features, structure and function of the (a) forest ecosystem (b) grassland ecosystem (c) desert ecosystem (d) aquatic ecosystems (ponds, rivers, oceans)													
Module II	ENVIRONMENT AL ISSUES AND SOLUTIONS										7 Hours		
Current Environmental Issues: Acid rain, Ozone layer depletion, Global warming, Green house effect. Solutions: 12 principles of green chemistry-Rain water harvesting.													
Module III	BIODIVERSITY										10 Hours		

Introduction to biodiversity -genetic, species and ecosystem diversity – value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values – hot-spots of biodiversity – threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – conservation of biodiversity: In-situ and ex-situ conservation of biodiversity.		
Module IV	NATURAL RESOURCES	10 Hours
Forest resources: Use and over-exploitation, deforestation- timber extraction, mining, dams and their effects on forests and tribal people – Water resources: Use and over utilization of surface and ground water, dams-benefits and problems – Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources – Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity– Energy resources: Growing energy needs, renewable and nonrenewable energy sources, use of alternate energy sources. Energy Conversion processes Biogas – production and uses, anaerobic digestion – Land resources: Land as a resource, land degradation, man induced landslides, soil erosion and desertification – role of an individual in conservation of natural resources – Equitable use of resources for sustainable lifestyles.		
Module V	ENVIRONMENTAL POLLUTION	10 Hours
Definition – Source, causes, effects and control measures of: (a) Air pollution (b) Water pollution(c) Soil pollution - soil waste management: causes, effects and control measures of municipal solid wastes – (d) Marine pollution (e) Noise pollution –(f) Nuclear pollution (g) Thermal pollution role of an individual in prevention of pollution.		
		TOTAL: 45 Hours
REFERENCES:		
1.Trivedi.R.K., "Handbook of Environmental Laws, Rules, Guidelines, Compliances and Standards", Vol. I and II, Enviro Media, 3rd edition, BPB publications, 2010.		
2.Cunningham, W.P. Cooper, T.H. Gorhani, "Environmental Encyclopedia", Jaico Publ., House, Mumbai, 2001.		
3.Dharmendra S. Sengar, "Environmental law", Prentice hall of India PVT LTD, New Delhi, 2007.		
4.Rajagopalan, R, "Environmental Studies-From Crisis to Cure", Oxford University Press, 2005.		
5.Benny Joseph, "Environmental Science and Engineering", Tata McGraw-Hill, New Delhi, 2006		
6. https://en.wikipedia.org/wiki/Carbon_capture_and_storage		
7.Ravikrishnan “Environmental Science and Engineering” Sri Krishna Hi-tech Publishing		

2302BM351	HUMAN ANATOMY AND PHYSIOLOGY LABORATORY	L	T	P	C
		0	0	2	1

PREREQUISITE :

Basic knowledge of biology

Course Objectives:

1. To identify all the organelles of an animal cell and their function
2. To understand structure and functions of the various types of systems of human body
3. To demonstrate their knowledge of importance of anatomical features and physiology of human systems

Course Outcomes:

After completion of the course, Student will be able to

- | | |
|-------------|---|
| CO1: | Perform and analyse hematological test (HB1,WBC) for diagnosis of blood related disorders |
| CO2: | Demonstrate immune hematology techniques and access transition compatibility |
| CO3: | Record and analyse biosignals (ECG,EMG)to diagnose cardio vascular and neuro vascular disorders |
| CO4: | Apply biomedical instrumentation to evaluate and classify sensory system disorders |
| CO5: | Interpret laboratory findings and correlate them with clinical conditions, ensuring documentation team wise and professional practice |

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	2	-	-	-	-	-	1	1
CO2	2	-	-	-	2	-	-	-	-	-	1	1
CO3	3	-	-	1	2	-	-	-	-	-	1	1
CO4	2	2	-	1	2	-	-	-	-	-	1	1
CO5	2	2	-	1	2	-	-	-	-	-	1	1

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO3
CO1	1	1	-
CO2	1	1	-
CO3	1	1	-
CO4	1	1	-
CO5	1	1	-

LIST OF EXPERIMENTS:

1. Diagnosis of platelet related disorders and hemophilia by analyzing bleeding and clotting time
2. Diagnosis of anemia by measuring the amount of hemoglobin in the blood.

3	Determine whether the donors blood group is compatible with the recipients blood group for a safe blood transfusion.
4.	Diagnosis of leukocytosis and leukopenia by counting the WBC present in the blood.
5.	Diagnosis of myocardial infarction from abnormal PQRST waveform
6.	Diagnosis of myopathies and neuropathies using EMG
7.	Diagnose the conductive and sensory-neural type of hearing loss.
8.	Identification of refractive errors like myopia , hyperopia and presbyopia.
Total :30 Hours	
References:	
1.Elaine.N. Marieb, "Essential of Human Anatomy and Physiology",12th Edition, Pearson Education, New Delhi,2018	
2.Gillian Pocock, Christopher D. Richards, "The Human Body - An Introduction for Biomedical and Health Sciences",Oxford University Press, USA,2009	
3.William F.Ganong, "Review of Medical Physiology", 26nd Edition, McGraw Hill, New Delhi, 2019	

2302BM352	CIRCUITS AND NETWORKS LABORATORY	L	T	P	C
		0	0	2	1

PREREQUISITE :

1.Electronics device lab

Course Objectives:

1. Be exposed to RL and RC circuits
2. Be familiar with Thevenin & Norton theorem KVL & KCL, and Super Position Theorems
3. To Know about series, parallel resonance circuits
4. To observe the characteristics of nodal analysis

Course Outcomes:

After completion of the course, Student will be able to
CO1: Design RL and RC circuits
CO2: Verify Thevenin & Norton theorem KVL & KCL, and Super Position Theorems
CO3: Have a better understanding of nodal analysis
CO4: Analyze different characteristics of R L C Circuits

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	1	-	-	-	1	-	1	-
CO2	3	2	2	1	1	-	-	-	1	-	1	-
CO3	3	2	2	1	1	-	-	-	1	-	1	-
CO4	3	2	2	1	1	-	-	-	1	-	1	-

COs Vs PSOs MAPPING

COs	PSO1	PSO2	PSO3
CO1	1	1	-
CO2	1	1	-
CO3	1	1	-
CO4	1	1	-

List of Experiments:

1. Measure the voltage across resistive circuit and verify it, using Kirchhoff's Voltage law (KVL)
2. Measure current in various branches of the given circuit and verify it, using Kirchhoff's current law (KCL)..
3. Measure current through and voltage across given branch of electric network and verify it by mesh analysis.
4. Measure voltage at particular node and current through branch of network and verify it by nodal analysis
5. Observe transient response of RL series circuit with DC excitation.
6. Observe transient response of RC series circuit with DC excitation
7. Measure current through given branch of network and voltage across given element of the circuit and verify it applying Superposition theorem
8. Measure short circuit current and Norton's resistance of the given circuit and verify it using Norton's theorem
9. Measure open circuit voltage and thevenin's resistance of the given circuit and verify it using Thevenin's theorem
10.Vary load resistance to transfer Maximum power in the given circuit using maximum power transfer theorem
11.Measure voltage and current in the given R-L .R-C,R-L-C series and parallel circuit and calculate

active, reactive and apparent power consumed in the circuit

References:

- 1.Muhammad H. Rashid, “Microelectronic Circuits: Analysis and Design”, Cengage Learning, 6th Edition, 2013.
- 2.Robert L. Boylestad, “Electronic Devices and Circuit Theory”, 11th Edition, 2015.
- 3.Robert B. Northrop, “Analysis and Application of Analog Electronic Circuits to Biomedical Instrumentation”, CRC Press, 2004.